

Argonia 2027

American Institute of Aeronautics and Astronautics
University of California, Merced



Propulsion Development & Rocketry
University of California, Merced

Overview

Argonia CDR

Length 138 in, max. diameter 3.12 in

Mass with no motors 8.6 kg

Mass with motors 13.2 kg

Stability: 6.32 cal / 14.2 %

CG: 81.395 in

CP: 101 in

at $M=0.300$



Flight configuration: [K250W-0; L1090W-0]

Apogee: 15402 ft

Max. velocity: 312 m/s (Mach 0.932)

Max. acceleration: 90.2 m/s²

Material Choices

Fiberglass

- Airframe, nosecone, fins, couplers, bulkheads,

Aluminum

- Nose tip, air brakes, boat tail, thrust plate,

Carbon Fiber

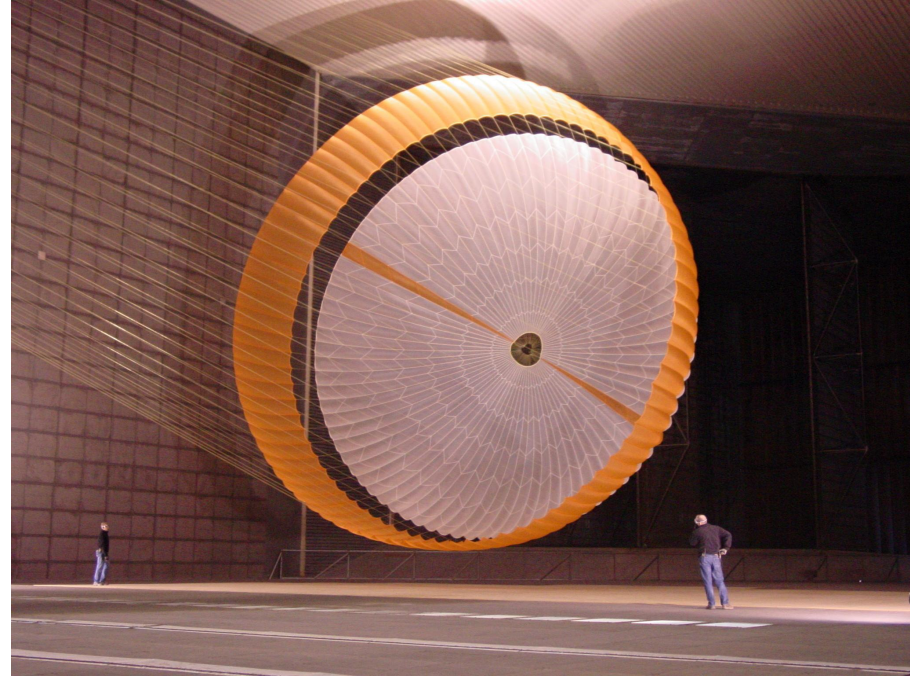
- Layup on fin cans

Dimensions of hardware will try to remain $\frac{1}{4}$ -20 and 4-40 for screws except for boat tail of 10-32



Recovery hardware choices

- Booster (touchdown = 8.42 m/s)
 - Drogue (diameter - 15")
 - Parachute Main (diameter - 45")
 - Sustainer (touchdown = 8.08 m/s)
 - Drogue (diameter - 15")
 - Parachute (diameter - 45")
 - $\frac{1}{4}$ in eyebolts, assuming 250 lbs jolt force
- Single separation dual Deploy for both stages.



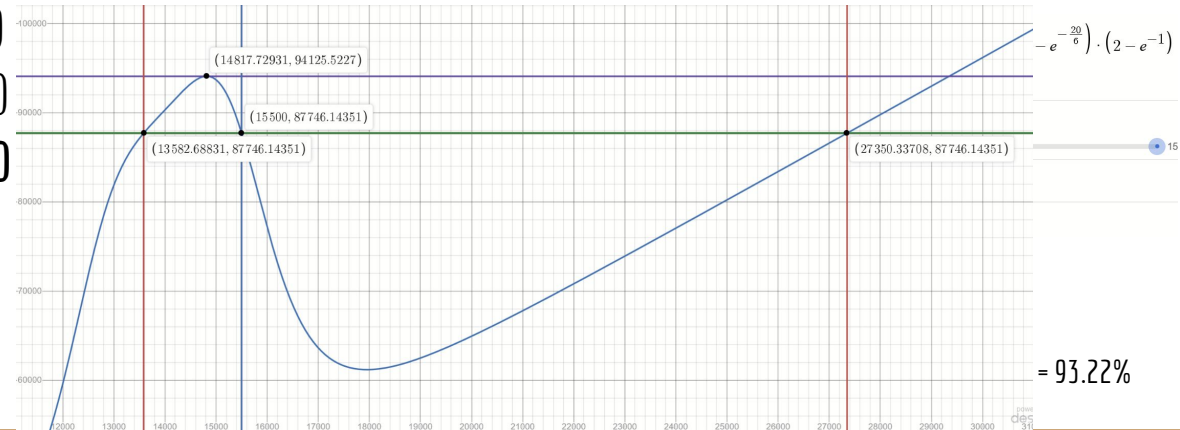
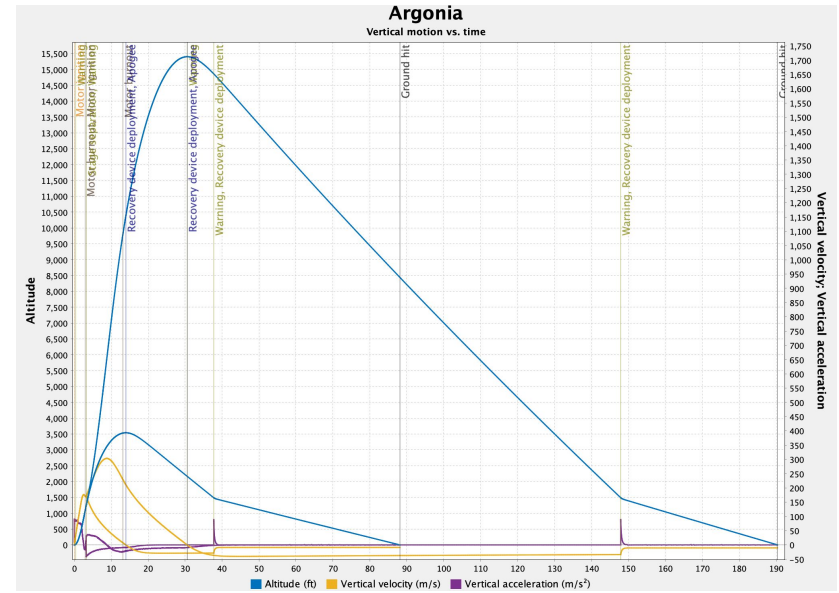
Mass Allocations (Subsystems)

- Avionics Coupler Total - 441 g per
- Air Brakes - 1 kg (budget)
- Boat tail 250g
- Thrust plate 150g

Component	Individual Weight (grams)	Quantity	Total Weight (grams)
Blue Raven	6.6	2	13.2
Blue Raven and Blue Raven Battery	13	4	52
Blue Jay	4.7	2	6.7
Featherweight GPS	15	2	30
Featherweight GPS battery	9	2	18
5 connector wago	4	2	8
2 connector wago	2.4	4	9.6
Wires	10	1	10
Pull switches	1.6	6	9.6
RunCam split 4 V2	10.2	1	10.2
Runcam split 4 V2 Battery	46	1	46
9in threaded rod	42	4	168
Nuts	4	36	144
Washers	2.3	16	36.8
Eyebolt	59	4	236
PLA Sled (10mm x 75mm x 9 inches)	42	2	84
Total Weight =			882.1

Projected apogee

- Targeting 14817 ft
- OpenRocket Projection - 15500 ft at 310 m/s
- Utilizing air brakes to control velocity → Apogee
- Maximum store for current configuration (before braking 29 kft) = 94,125
- Score at target + 500ft = 90,758 (96%)
- Score at target - 500ft = 92,286 (98%)
- **Current score = 87,746 (93.2% yield)**



Air Brakes

Mechanical :

- CNC manufactured 6061-T6 Al
- Weight budget of 2.2lbs
- 2.875" diameter
- Servo with central linkage actuation

Electrical :

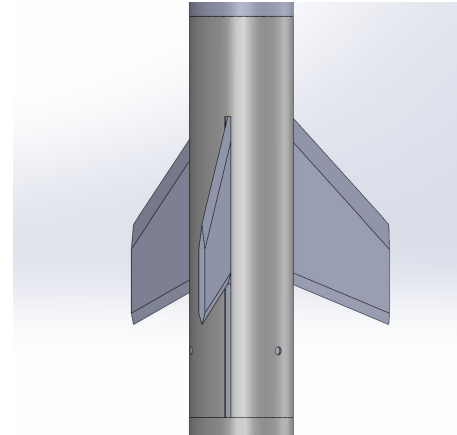
- 4 layer PCB board (signal / ground / power / signal)
- ESP32-WROOM (MCU), MS5600 (Barometer), MPU(6050), Sparkfun KX134
- Module breakout board style
- SPI/I2C communication



(Adapt Previous AB)

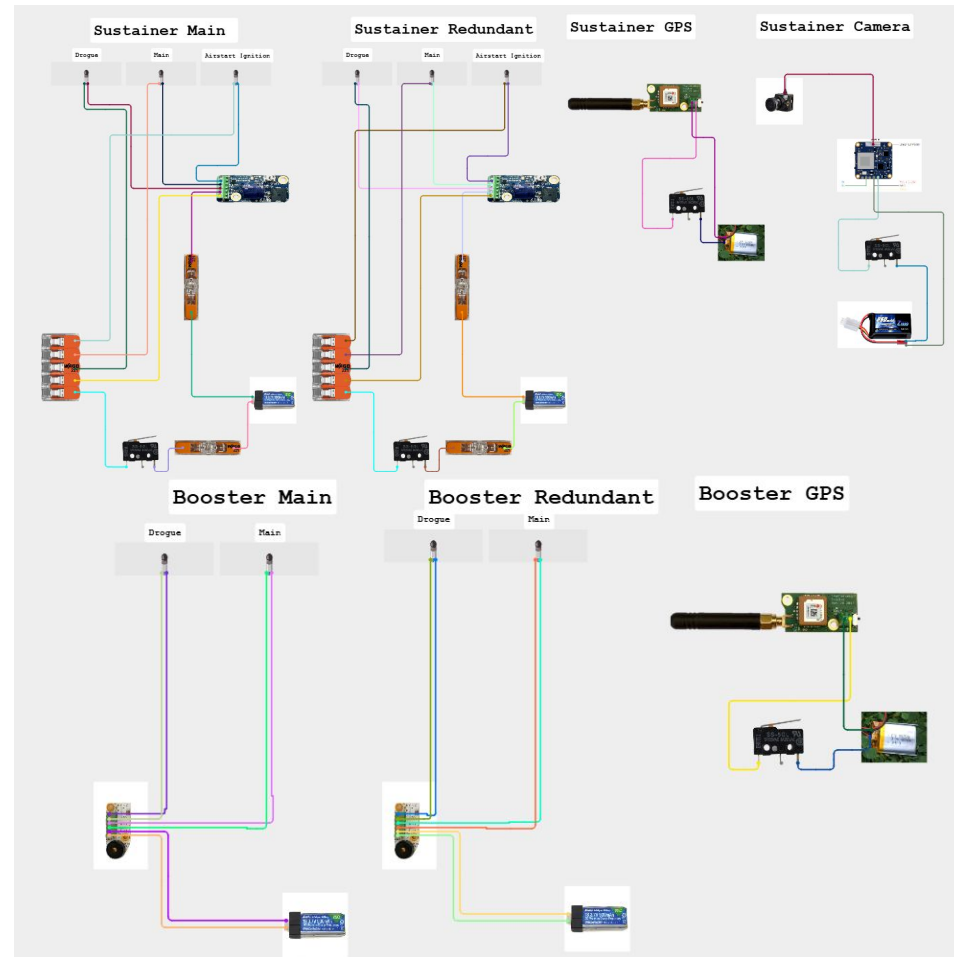
Manufacturing

- Fins
 - Fiberglass sheets shaped with dremel to match fin dimension with bevels
- Boat Tail & Thrust Plate
 - Machined by Machine Shop, Made of 6061 t6 aluminum, Thrust plate is threaded and will be bonded to the airframe with epoxy, the threading on the thrust plate is to fasten the boat tail on to it.
- Fin Cans
 - 2 centering rings, mmt, fincan coupler, and the fins will all be bonded with epoxy
 - 3 lamina carbon fiber layups on the fins
 - Sustainer fincan fin slots will be filled with fiberglass
- Nose cone
 - Mandrill will be used, 6 fiberglass sleeves, mold release



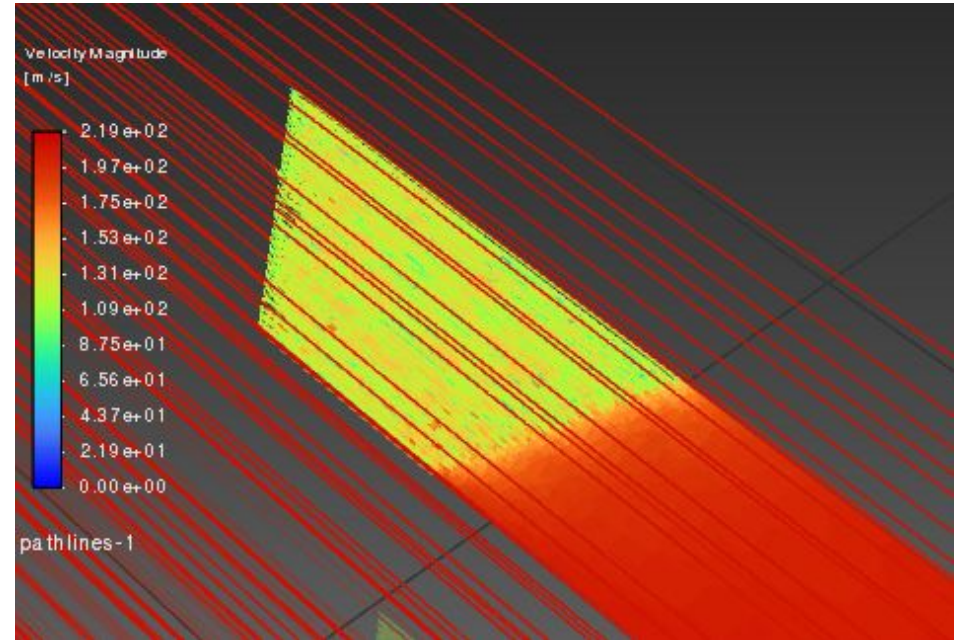
Avionics flight logic

- Single separation dual deploy configuration for both the sustainer and booster
- Stage; 2 BlueRaven FCs
- Sustainer; 2 BlueJay FCs
- Featherweight GPS for both Stage and Sustainer
- Altimeters built-in to FCs
- RunCam Split 4 V2 for flight footage
- Utilize electronics for all electronics, except Blue Jay which uses magnetic switches



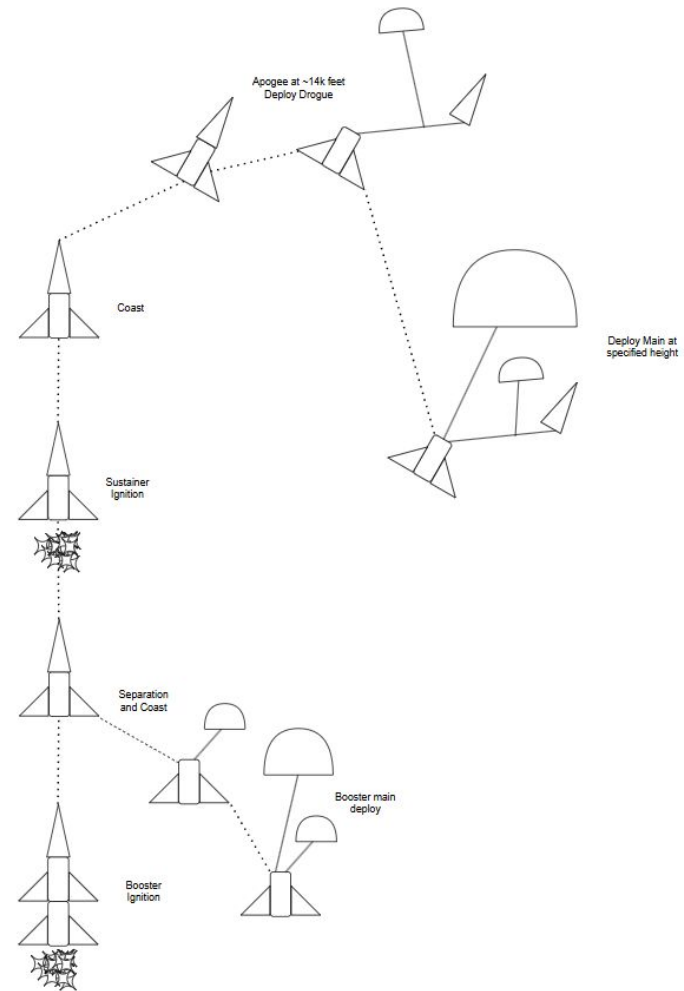
Simulations

- Confirming Openrocket simulation assumptions
 - Confirmed Openrocket fin C_d estimates are conservative
 - Ansys simulated C_d 0.4, openrocket 0.5
 - Confirmed Openrocket fin C_d estimates use wing area as reference area, not cross-section
 - Achieving residuals in $1e-5$ range
- Building knowledge within the club
 - Documenting Fluent and Mechanical workflows in club-wide papers
 - Introducing new members to simulation workflows



Staging

- Pyrotechnic Staging
- Shear pins
- Motor retained via forward motor retention and aluminum tape



Closing comments

- Ordering parts
 - McMaster-Carr
 - Wildman
 - Aerotech
 - JLCPCB